The D. G. Ruparel College
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TITLE :

Aim:- Interface Fingerprint sensor and 16x2 LCD with Raspberry and display message RFID UID matched or unmatched.

* Hardware Requirements:-
1. Raspberry Pi Model
2. Jumper Wire (Female
3. 16x2 character LCD with soldered HW-61(PCF8574) I2C Backpack.
4. R307/R305 Optical Fingerprint sensor Module.

* Pinout & Interfacing Connection Table:-

Device/ Module	Module Pin	Pin Function	Raspberry Pi Pin Name	Physical Pin No.
	GND	Ground Reference	GND	Pin 6
16x2 I2C LCD (HW-61 Backpack)	VCC	Supply Voltage(+5V)	5V Power	Pin 2
	SDL	I2C Serial Data	GPIO 2 (I2C1 SDA)	Pin 3
	SCL	I2C Serial Clock	GPIO 3 (I2C1 SCL)	Pin 5

R307/ R305 Fingerprint Sensor	Red (VCC)	Module Power (+5V)	5V Power	Pin 4
	Black (GND)	Ground Reference	GND	Pin 9/ 14
	Yellow/ Green (TXD)	UART Transmit Data	GPIO 15 (UART RXD)	Pin 10
	White/Blue (RXD)	UART Receive Data	GPIO 14 (UART TXD)	Pin 8

* Software Installation

1. Enable Interfaces:- Run sudo raspi-config. Under Interface Options, enable I2C. Under Serial Port, select No for login shell and Yes for hardware serial. Reboot with sudo reboot.

2. Install Dependencies:-

- sudo apt-get update && sudo apt-get install -y i2c-tools python3-pip python3-serial

- pip3 install RPLCD adafruit-circuitpython-fingerprint smbus2

3. Verify I2C Address: Run i2cdetect -y 1 (Address defaults to 0x27 or 0x3F).

*Code:-

A. Enrollment script (enroll.py)

```
import time, serial, adafruit_fingerprint

uart = serial.Serial("/dev/serial0", baudrate=57600,
 timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint
                                    (uart)

location = int(input("Enter Slot ID (1-127): "))

print("Place finger on sensor...")
while finger.get_image() != adafruit_fingerprint.
                                    OK:
    pass
finger.image_2_tz(1)

print("Remove finger...")
time.sleep(2)
while finger.get_image() == adafruit_fingerprint.
                                    OK:
  pass

print("Place SAME finger again...")
while finger.get_image() != adafruit_fingerprint.
```

```
            .OK:
        pass
finger.image_2_tz(2)

if finger.create_model() == adafruit_fingerprint.
                                OK:
    if finger.store_model(location) ==
adafruit_fingerprint.OK:
        print(f"Enrolled successfully at slot
{location}!")
    else:
        print("Error saving template.")
else:
    print("Fingerprint mismatch. Try again!")
```

B. Verification System (main_verify.py)

```
import time, serial
from RPLCD.i2c import CharLCD
import adafruit_fingerprint

lcd = CharLCD('PCF8574', 0x27, cols=16, rows=2)
uart = serial.Serial("/dev/serial0", baudrate=
                                    57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint
                                    (uart)

def show(l1, l2=""):
    lcd.clear()
    lcd.cursor_pos = (0, 0)
    lcd.write_string(l1)
    lcd.cursor_pos = (1, 0)
    lcd.write_string(l2)

show("System Ready", "Place Finger")

while True:
  if finger.get_image() == adafruit_fingerprint.
                                    OK:
    show("Processing....", "Please wait")
    if finger.image_2_tz(1) ==
```

```
adafruit_fingerprint.OK:
        if finger.finger_search() ==
adafruit_fingerprint.OK:
            show("RFID UID MATCHED", f"ID:
{finger.finger_id}")
            time.sleep(3)
        else:
            show("RFID UID", "UNMATCHED")
            time.sleep(1.5)
            show("Access Denied", "")
            time.sleep(2)
    show("Place Finger", "")
time.sleep(0.1)
```

* Step-By-Step

1. Power off the Raspberry Pi 4 and wire all modules according to the pin connection table.
2. Power on Raspberry Pi and confirm I2C bus detection using i2cdetect -y 1.
3. Run python3 enroll.py in the terminal and follow on-screen prompts to register sample fingerprint into slot ID 1, 2, etc.
4. Launch the verification terminal: python python3 main_verify.py.
5. Test with enrolled (verify "RFID UID MATCHED") and unenrolled finger (verify "RFID UID UNMATCHED" & Access Denied.